Chapter 5

Limits and continuity

The concepts of limits and continuity lie at the very foundation of mathematical analysis. They formalize the intuitive ideas of *approaching* and *connectedness*, and provide the rigorous language needed to define derivatives and integrals.

Historically, these ideas evolved over centuries. Ancient Greek mathematicians like Eudoxus and Archimedes used rudimentary forms of limits in the *method of exhaustion* to compute areas and volumes. However, it was not until the 17th century, with the development of calculus by Newton and Leibniz, that the notion of a limit became central—though it remained informal and often controversial.

In the 19th century, mathematicians such as Cauchy and Weierstrass finally placed limits on a firm logical foundation using the famous ε-δ definitions. The key insight was to quantify the idea of "arbitrarily close" using two positive numbers: ε for the error tolerance in the function's output, and δ for the corresponding distance in the input. This rigorous framework eliminated the paradoxes and ambiguities that had plagued early calculus and remains the standard today.

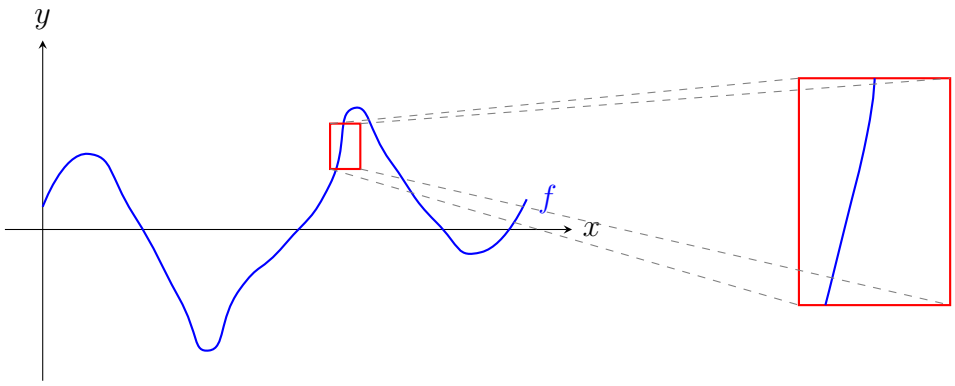


Figure 5.1: We will develop rigorous tools for "zooming-in"

5.1 Neighborhoods

To discuss continuity and limits, we need to be able to "zoom" into the graph of a function. To do this, we need a definition of a small subset around a given point that we want to investigate. This is called a *neighborhood*:

Neighborhood

Let $x_0 \in \mathbb{R}$ be a point on the real line and let $r > 0$ be a positive parameter. The **neighborhood of x_0 of radius** r is the *open* and bounded interval

$$I_r(x_0) = (x_0 - r, x_0 + r).$$

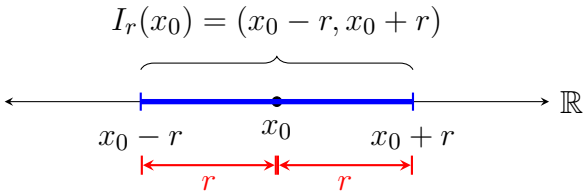


Figure 5.2: Neighborhood of x_0 with radius r

The idea will be that to understand what is happening at x_0, we shall take smaller and smaller neighborhoods of it, so that we can zoom in more and more. Moreover, it can happen that at the point x_0 itself our function is problematic (for example, maybe the value of the function is not defined there). For this reason we might want to look at a neighborhood that *excludes* the point x_0 itself:

Deleted neighborhood

Let $x_0 \in \mathbb{R}$ be a point on the real line and let $r > 0$ be a positive parameter. The **deleted neighborhood of x_0 of radius** r is the *open* and bounded union of intervals

$$I_r^*(x_0) = (x_0 - r, x_0) \cup (x_0, x_0 + r) = I_r(x_0) \setminus \{x_0\}$$

Note that x_0 does *not* belong to this set.

5.1.1 Neighborhoods of infinity

The same idea extends naturally to $+\infty$ and $-\infty$: instead of a radius r around a finite point, we use a threshold $M > 0$ beyond which points are considered "close to infinity".

Neighborhood of infinity

Let $M > 0$ be a positive parameter. The **neighborhood of $+\infty$ of threshold** M and the **neighborhood of $-\infty$ of threshold** M are the half lines

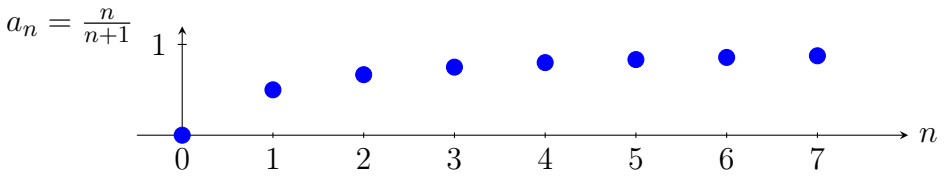
$$I_M(+\infty) = (M, +\infty), \qquad I_M(-\infty) = (-\infty, -M).$$

Since $+\infty$ and $-\infty$ are not themselves real numbers, there is no point to exclude: unlike the finite case, no separate deleted neighborhood definition is needed for them.

5.2 Limits of sequences

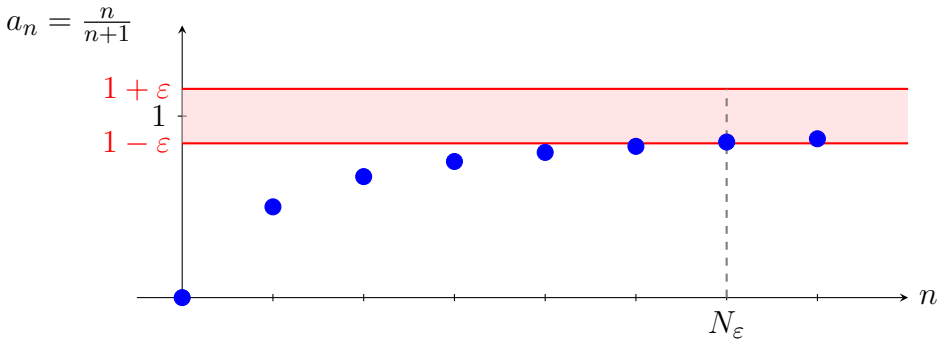
We start by considering sequences. Consider the following two examples which we've seen before:

Example 5.1: Consider the sequence $a_n = \frac{n}{n+1}$.



It is evident that this sequence 'approaches' the number 1. However, we want to make this rigorous. To do this, we look at a small neighborhood of 1 and ask whether the sequence eventually enters this neighborhood (and never leaves!). So our question is this:

> **Q:** *For any $\varepsilon > 0$, does there exist $N_\varepsilon \in \mathbb{N}$ such that for all $n > N_\varepsilon$, $a_n \in I_\varepsilon(1) = (1-\varepsilon, 1+\varepsilon)$? Or, equivalently, is it true that*
>
> $$|1 - a_n| < \varepsilon, \qquad \forall n > N_\varepsilon?$$



So here's the process:

- Fix $\varepsilon > 0$ ***(think of ε as being positive but VERY small!)***
- Compute $1 - a_n$:

$$1 - a_n = 1 - \frac{n}{n+1} = \frac{n+1-n}{n+1} = \frac{1}{n+1}.$$

- Apply the condition $|1 - a_n| < \varepsilon$:

$$\frac{1}{n+1} < \varepsilon \quad \Leftrightarrow \quad 1 < \varepsilon(n+1) \quad \Leftrightarrow \quad \frac{1}{\varepsilon} < n+1$$

- Find N_ε: define

$$N_\varepsilon = \left\lceil \frac{1}{\varepsilon} \right\rceil.$$

- Verify the condition for $\forall n > N_\varepsilon$:

$$\forall n > N_\varepsilon, \qquad n+1 > N_\varepsilon + 1 = \left\lceil \frac{1}{\varepsilon} \right\rceil + 1 \geq \frac{1}{\varepsilon} + 1 > \frac{1}{\varepsilon}$$
$$\Leftrightarrow \qquad \frac{1}{n+1} < \varepsilon$$

We have therefore answered the previous question affirmatively:

A: *Yes,*

$$\forall \varepsilon > 0,\ \exists N_\varepsilon \in \mathbb{N},\ \text{s.t. } \forall n > N_\varepsilon,\ |1 - a_n| < \varepsilon.$$

In fact, there is an explicit choice for N_ε, namely: $N_\varepsilon = \lceil \frac{1}{\varepsilon} \rceil$.

Since this is true *for every* $\varepsilon > 0$, the sequence $a_n = \frac{n}{n+1}$ gets arbitrarily close to the number 1. We say that 1 is the *limit* of this sequence.

Finite limit of a sequence

Let $\{a_n\}_{n\in\mathbb{N}}$ be a real-valued sequence. We say that the sequence **converges to** $\ell \in \mathbb{R}$ if for any $\varepsilon > 0$ there exists $N_\varepsilon \in \mathbb{N}$ such that for any $n > N_\varepsilon$, we have $|\ell - a_n| < \varepsilon$ (i.e. $a_n \in I_\varepsilon(\ell)$). In this case we write

$$\lim_{n\to\infty} a_n = \ell.$$

The condition for convergence can be written symbolically as:

$$\forall \varepsilon > 0,\ \exists N_\varepsilon \in \mathbb{N},\ \text{s.t. } \forall n > N_\varepsilon,\ |\ell - a_n| < \varepsilon.$$

Returning to Example 5.1, we conclude that:

$$\lim_{n\to\infty} \frac{n}{n+1} = 1.$$

A sequence might also grow indefinitely and 'tend' to $+\infty$ (for example, $a_n = e^n$), or decrease and 'tend' to $-\infty$ (for example, $a_n = -n^8$). This can be formalized as follows:

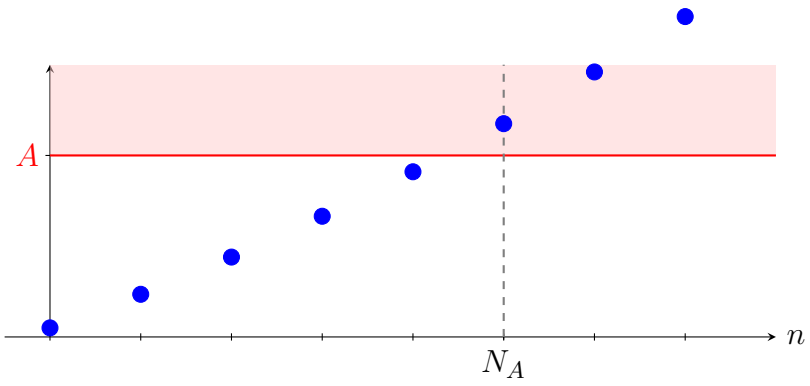
Divergent sequence (to $+\infty$)

Let $\{a_n\}_{n\in\mathbb{N}}$ be a real-valued sequence. We say that the sequence **diverges to** $+\infty$ if for any $A > 0$ there exists $N_A \in \mathbb{N}$ such that for any $n > N_A$, we have $a_n > A$ (i.e. $a_n \in I_A(+\infty)$). In this case we write

$$\lim_{n\to\infty} a_n = +\infty.$$

The condition for convergence can be written symbolically as:

$$\forall A > 0,\ \exists N_A \in \mathbb{N},\ \text{s.t. } \forall n > N_A,\ a_n > A.$$



Here we ***think of A as being VERY big!***

We can make a similar definition for a sequence divergent to $-\infty$:

Divergent sequence (to $-\infty$)

Let $\{a_n\}_{n\in\mathbb{N}}$ be a real-valued sequence. We say that the sequence **diverges to $-\infty$** if for any $A > 0$ there exists $N_A \in \mathbb{N}$ such that for any $n > N_A$, we have $a_n < -A$ (i.e. $a_n \in I_A(-\infty)$). In this case we write

$$\lim_{n\to\infty} a_n = -\infty.$$

The condition for convergence can be written symbolically as:

$$\forall A > 0,\ \exists N_A \in \mathbb{N},\ \text{s.t. } \forall n > N_A,\ a_n < -A.$$

Here too, we ***think of A as being VERY big!***

Example 5.2: Consider the sequence $a_n = \frac{n!}{n^{100}}$. Does $\{a_n\}_{n\in\mathbb{N}}$ have a limit as $n \to \infty$? If so, what is it? Does it diverge?

Solution. Let us rewrite the nth term as follows, taking n to be very large (more than 100):

$$\begin{aligned} a_n &= \frac{n!}{n^{100}} \\ &= \frac{n\cdot(n-1)\cdot(n-2)\cdots(n-98)\cdot(n-99)\cdot(n-100)\cdots 3\cdot 2\cdot 1}{n\cdot n\cdots n} \\ &= \frac{n\cdot(n-1)\cdot(n-2)\cdots(n-98)\cdot(n-99)}{n\cdot n\cdots n}\cdot(n-100)\cdots 3\cdot 2\cdot 1. \end{aligned}$$

Observe that

$$\forall n > 198, \qquad \frac{n-99}{n} > \frac{1}{2}.$$

Hence, for all $n > 198$:

$$\begin{aligned} a_n &= \frac{n}{n} \cdot \frac{n-1}{n} \cdots \frac{n-98}{n} \cdot \frac{n-99}{n} \cdot (n-100)! \\ &> \frac{1}{2} \cdot \frac{1}{2} \cdots \frac{1}{2} \cdot \frac{1}{2} \cdot (n-100)! \\ &= \frac{1}{2^{100}} \cdot (n-100)! \end{aligned}$$

Now, $\frac{1}{2^{100}}$ is some (*very* small) positive number, however $(n-100)!$ diverges as $n \to \infty$, so that eventually, for any $A > 0$, there exists $N_A \in \mathbb{N}$ such that $\frac{1}{2^{100}} \cdot (n-100)! > A$ for all $n > N_A$. Hence

$$\lim_{n\to\infty} a_n = +\infty.$$

Indeterminate sequences

Some sequences do not converge but also do not diverge. These are called **indeterminate sequences**. Examples include:

- The sequence $a_n = (-1)^n$ is *bounded* but does *not* converge.
- The sequence $a_n = n(-1)^n$ is *not* bounded (not from below and not from above) *and* does *not* converge.

A crucial feature of indeterminate sequences is that their values are not monotone: they increase and decrease repeatedly. The following theorem demonstrates this.

Theorem 5.1 (Monotone Convergence Theorem)**:** Let $\{a_n\}_{n\in\mathbb{N}}$ be a real-valued sequence. Assume that there exists $N \in \mathbb{N}$ such that for all $n > N$, the sequence is monotone. Then the sequence *cannot* be indeterminate. More precisely:

- In the case that the sequence is monotone increasing:
  - If $\sup\{a_n \mid n > N\} < +\infty$, then $\lim_{n\to\infty} a_n = \sup\{a_n \mid n > N\}$.
  - If $\sup\{a_n \mid n > N\} = +\infty$, then the sequence diverges to $+\infty$.
- In the case that the sequence is monotone decreasing:
  - If $\inf\{a_n \mid n > N\} > -\infty$, then $\lim_{n\to\infty} a_n = \inf\{a_n \mid n > N\}$.
  - If $\inf\{a_n \mid n > N\} = -\infty$, then the sequence diverges to $-\infty$.

Proof. We only prove for the monotone *increasing* case (the decreasing case follows the same proof). For brevity we shall write

$$\sup_{n>N} a_n = \sup\{a_n \mid n > N\}$$

$\star$ Suppose that $\sup_{n>N} a_n = \ell < +\infty$. Fix some $\varepsilon > 0$. By the definition of the supremum,